

CS 4423 Game Development

Final Project Rubric

Team Size: 1

Objective

Create a game.

Due Dates

3/31/26 - Prototype and Progress Report

- Turn in a document on canvas, along with a recording of your video, and a link to access your code (github, zipfile, google drive, etc.)

5/5/26 - Final Submission

- Turn in a document on canvas along with a recording, and a link to access your code (github, zipfile, google drive, etc.)

Following Guides is great! But if you follow an online guide or video word for word, I will not accept your project. Also don't try to submit a game you found off the internet.

Grading (100 points total)

I reserve the right to award points if I like what the student made but they somehow strayed outside the bounds of the rubric.

The following pages describe the grading for your final submission.

Final Writeup & Video (required)

I will include a template for your final writeup that you can fill out. Underlined sections are extra important. This will ask for...

- The Name of your game.
- A Youtube link to a gameplay video of your project. Please record a few minutes of footage and show me everything your game has to offer.
- **A list of all assets you used from the internet and where they specifically came from. (include links)**
- A brief summary of your game, similar to what was included in your design document.
- How to play your game (controls, goal, etc.)
- Requirements you completed listed with point values and their summed total.
- The bounties you completed listed with point values and their summed total.
- Additional features you completed you want me to be aware of.

I will provide a template for the writeup and self grading.

Requirements (10 points total)

To receive points from this category, you must complete the following.

- Start menu with the name of your game, play and quit button. **(2 points)**
- Dynamic audio that plays in response to player actions or events on screen (think of point pickups from the midterm) **(3 points)**
- Some form of player input **(5 points)**

Subjective Criteria (50 points)

These categories are scored based on my judgment of your end product.

Coherency (10 points)

- Does your game feel like a completed game, or random requirements you threw together from the rubric?
- Are there non-trivial things for the player to do?
- Does a clear gameplay loop exist (if appropriate, not all games will need this)
- Does the game appear to have incomplete features, or needed features that are clearly missing?

Performance & Stability (10 points)

- Receive up to 10 points based on the following
 - Does the game run at a consistent 60fps or higher?
 - Does the game run without crashing?
 - Are there obvious bugs while playing?

Ambition / Accomplishments (20 points)

- How does the amount of work you did compare with that of your classmates?
- Does it feel as though you made good use of your time?
- New Grading Thought experiment : 5 points per class project unit of work in the project.
- 4x as much work means a perfect score.

Visual Cohesion (10 points)

- Are you relying on default assets within Unity? Or creating your own?
- Did you develop an overall look for your project.

Bounties (40 points total)

Select from the following to receive up to 40 points of credit. You don't need to do all of these. You don't need to do most of these. Pick and choose the ones you're most interested in.

You can receive 40 points from this category, but pursuing additional bounties will be considered in subjective categories.

Each bounty is matched with a point value. A flat point value indicates that I will award full credit for the feature unless it is not working properly. Score values labeled **up to X** points mean that I will be grading based on the complexity of your implementation.

AI

- Include any form of AI, even simple with predefined behavior. **(2 points)**
- Upgrade the AI to use a state machine, allowing for multiple behaviors. **(4 points)**
- Enable the AI to pathfind through the environment. Using A* or similar algorithm **(6 points)**
- Allow AI to interact with each other, not just the player. **(3 points)**
- Include additional varieties of AI with unique behaviors **(2 points each, maximum 3 extra AIs)**
- If your AI is playing a board game / card game / management game **(3 points)**

Levels

- Include at least 3 separate areas / maps / zones the player can visit. **(2 points)**
- Build a game with a metroidvania style map, allowing players to return to previous locations. **(5 points)**
- The main menu does not count as a level! **(0 points)**

Level Design

- Make use of moving terrain in your scene, such as elevators, moving platforms. **(3 points)**
- Make use of interactable objects in your scene that change state based on player interaction, such as doors, levers, or buttons. **(3 points)**
- Include destructible terrain. **(3 points)**
- Include environmental hazards **(2 points)**
- Include colliders with beneficial effects **(2 points)**
- Include checkpoints that the player returns to on failure. **(2 points)**

Procedural Generation

- Procedurally generate environments for the player. **(up to 10 points)**
- Generate objects with random parameters. **(up to 5 points)**

Particle Systems

- Create your own unique effects using particle systems. **(up to 6 points)**

Shaders

- Create your own complex shaders with Unity's shader graph, or writing HLSL **(up to 10 points)**

VFX

- Create your own visual effects using unity's VFX graph. **(up to 10 points)**

User Interface

- Change default Unity text and texture options for your UI elements. **(2 points)**
- Add a custom image for the mouse cursor **(1 point)**
- Include UI elements utilizing the vertical, horizontal, or grid layout groups. **(4 points)**
- Include draggable UI elements that the player can move with their mouse. **(5 points)**

Character Customization

- Include a character creator in your game **(up to 8 points)**
- Include 2 or more playable characters in your game **(4 points)**
- Have your character change in appearance over the course of the game **(2 points)**

- Add a skill tree or skill unlock system that gives the player different options on how to play **(up to 8 points)**

Physics

- Make use of Unity's cloth physics **(2 points)**
- Allow the user to launch projectiles of some sort **(2 points)**
- Allow the player to fire projectiles in an arc, and provide a visual of where they will go **(4 points)**
- Create a gravity system that pulls objects toward one another or simulates orbits **(3 points)**
- Make use of ragdolls **(4 points)**

Puzzle

- Create a unique puzzle mechanic based around a specific / mostly original idea. **(up to 8 points)**

Difficulty

- Provide difficulty options to the player. **(3 points)**
- Add dynamic difficulty, making the game easier or harder depending on player performance **(up to 6 points)**

Advanced Audio

- Require the player to match their actions to the music in some way. **(up to 6 points)**
- Include additional audio sliders categories **(1 points per slider, max 2 additional sliders)**
- Add dynamic music, where different effects or instruments fade in and out depending on player actions **(3 points)**.

Time

- Allow the user to pause the game **(2 points)**
- Make use of time dilation. **(2 points)**
- Include assets that change over time. **(up to 6 points)**
 - Examples: plants growing, animals getting hungry.
- Add a speedrunning timer to your game **(2 points)**
- Implement time travel **(8 points)**

Items

- Include different items that the player can equip and use. **(Up to 6 points)**
 - Food
 - Tools
 - Weapons
 - Fish food shaker.
- Allow the player to store these items in an inventory. **(5 points)**
- Make the inventory grid based so items slot in like tetris pieces **(8 points)**

Saving and Loading

- Allow the player to resume progress after quitting. **(5 points)**
- Include multiple save slots the player can select from. **(5 points)**

Multiplatform

- Make your project playable on a non-PC device such as phone or tablet. **(8 points)**
- Allow the player to use a handheld controller to play the game. **(4 points)**

Tutorial

- Include a tutorial sequence teaching the player how to play your game. **(up to 5 points)**

VR

- Create a game playable on a VR headset **(8 points)**
- Allow the player to play either in VR or on a desktop display **(4 points)**

Recording

- Allow the player to record audio into the game somehow **(4 points)**
- Add voice commands to the game **(up to 8 points)**

Photography

- Allow the player to save screenshots or images from the game somehow **(4 points)**
- Create a fully featured photo mode where the player can fly around and snap screenshots of whatever they want **(up to 8 points)**

Advanced Options

- Include a rebindable keys menu so that the player can rebind their keyboard inputs. **(6 points)**
- Provide options to change the look of your game. **(up to 5 points)**
 - Colors
 - Effects
 - Quality
 - Whatever you can think of!

Post Processing Effects

- Make use of Unity's post processing stack to change the overall look of your game. **(up to 5 points)**
- Create a unique post processing effect **(8 points)** **Note: I do not know how to do this but it is probably possible.**

Lighting

- Include baked lighting in your project **(4 points)**
- Create a day/night cycle for your game **(5 points)**

3D Modeling

- Include custom 3D models in your game **(up to 8 points)**
- Texture your models to elevate their appearance **(up to 6 points)**

Animation

- Include animated assets in your game **(up to 8 points, hand-made animations scored higher)**
- Utilize procedural animation in your project **(5 points)**

Environment

- Create an environment using Unity's terrain system **(up to 5 points)**
- Add weather effects like snow or rain to your game **(3 points)**

Meta

- Add in achievements that the player can unlock and view in a menu **(up to 5 points)**
- Add in a console that the player can type cheat codes or custom commands into **(3 points)**

Narrative

- Include a dialogue sequence. **(1 point)**
- Include branching dialogue sequences. Player responds, NPC dialogue changes based on choice. **(4 point)**
- Include a cutscene. **(up to 5 points)**
- Include voice acting. **(2 points)**
- Allow the player to read descriptions of items or objects they pick up / encounter **(3 points)**

Multiplayer

- Include networked multiplayer **(18 points)**
- Include local multiplayer **(9 points)**
- **Both of these are extremely difficult, don't do them unless you are absolutely dying to. We will not be covering networking in this class.**

Options

- Include an options menu that allows the player to.
 - Change the screen resolution **(3 points)**
- Include additional options in your required options menu
 - Vsync on and off **(2 pts)**
 - Fullscreen on and off **(2 pts)**
 - Visual settings such as brightness, contrast, coloration **(up to 6 points)**

Camera

- Make use of a smooth follow camera **(1 point)**
- Allow the player to move their point of view back and forth between multiple characters, multiple zones, etc. **(4 points)**
- Create a game with multiple fixed camera angles **(up to 8 points)**
- Make use of overlay cameras **(2 points)**

Create your Own Bounty

- I'm very open to ideas!
- Message me on canvas.